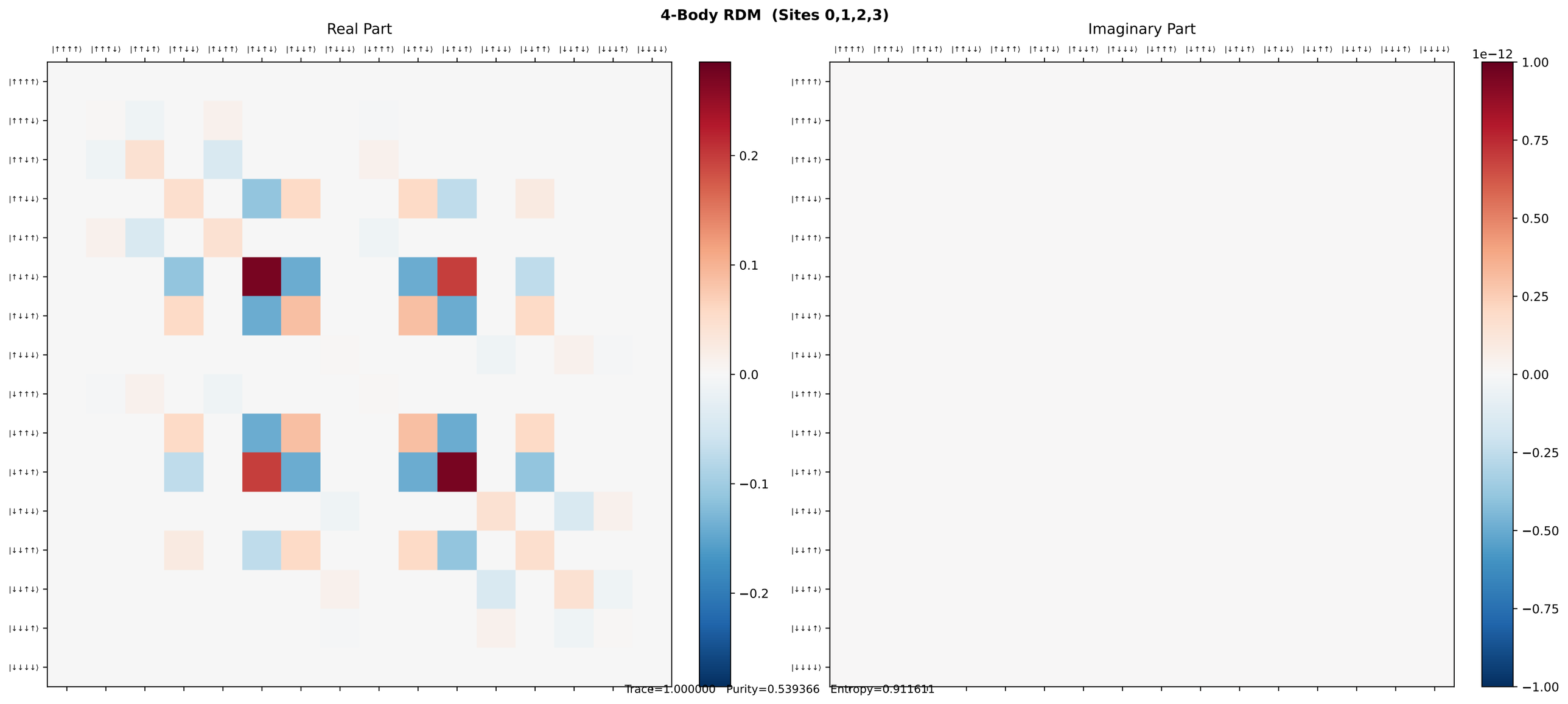
Genuine Multipartite Entanglement Analysis

```
Model                              Ground-state properties           GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E₀                : -0.04292319  Extracted n-spin RDM : n = 4
                                   E₁                : -0.03609210
System parameters                                                   Purity:
-----------------                  ⟨S²⟩              : -0.000000      γ = Tr(ρ²)
Sites (L)          : 6             S                 : -0.0000
Electrons (Ne)     : 6             ⟨Sz⟩              : 0.000000     Separable cut criterion:
Hopping (t)        : 0.1           Multiplicity 2S+1 : 1.0            γ_A = 1  ↔  ρ = ρ_A ⊗ ρ_B
Interaction (U)    : 4.0           Spin state        : Singlet (S=0)
Mode               : heisenberg                                     C_GME (Troiani et al.):
n-body order       : 4             Double occupancy                   C_GME = √(2 · min_A (1 − Tr(ρ_A²)))
                                   ----------------
                                   ⟨D⟩ total         : 0.010679     Decision logic:
                                   ⟨D⟩ / site        : 0.001780       γ = 1  and  all Tr(ρ_A²) < 1   → GME
                                                                      γ = 1  and  some Tr(ρ_A²) = 1  → No GME
                                                                      γ < 1                          → Mixed (convex-roof needed)

                                                                    Date: 2026-03-04  15:07
```


4-Body RDM (Sites 0,1,2,3)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.539366 Entropy=0.911611

```
RDM Metrics  (Sites = 0,1,2,3)
=========================================================

  Trace(ρ)                : 1.0000000000
  Purity Tr(ρ²)           : 0.5393659101
  Von Neumann entropy     : 0.9116106080
  Eigenvalue  min         : -2.471086e-17
  Eigenvalue  max         : 7.159102e-01
```

GME Analysis — 4-Body RDM (Sites 0,1,2,3)

Bipartitions (Sites 0,1,2,3)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.53936591$

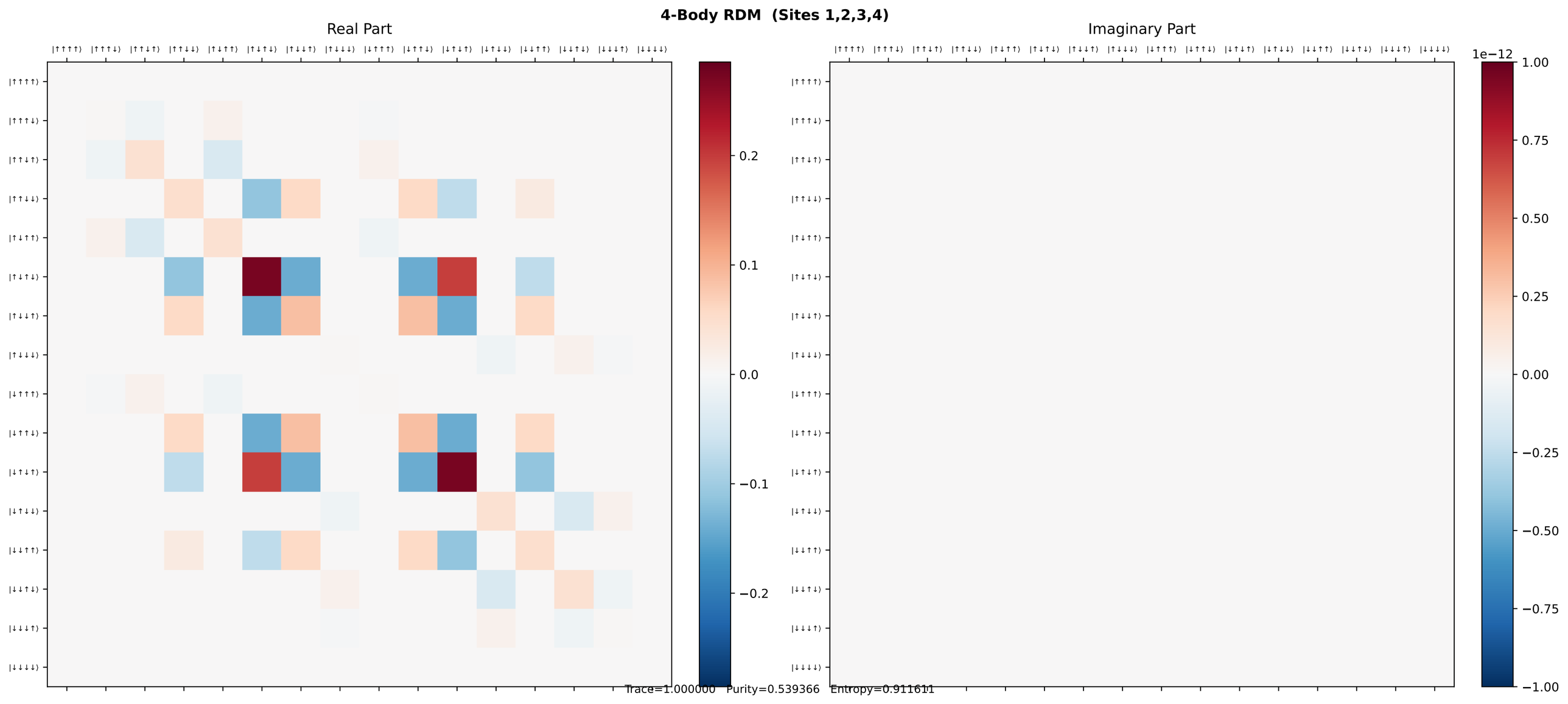
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 1,2,3,4)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.539366 Entropy=0.911611

```
RDM Metrics  (Sites = 1,2,3,4)
=========================================================

  Trace(ρ)                : 1.0000000000
  Purity Tr(ρ²)           : 0.5393659101
  Von Neumann entropy     : 0.9116106080
  Eigenvalue  min         : -2.449343e-17
  Eigenvalue  max         : 7.159102e-01
```

GME Analysis — 4-Body RDM (Sites 1,2,3,4)

Bipartitions (Sites 1,2,3,4)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.53936591$

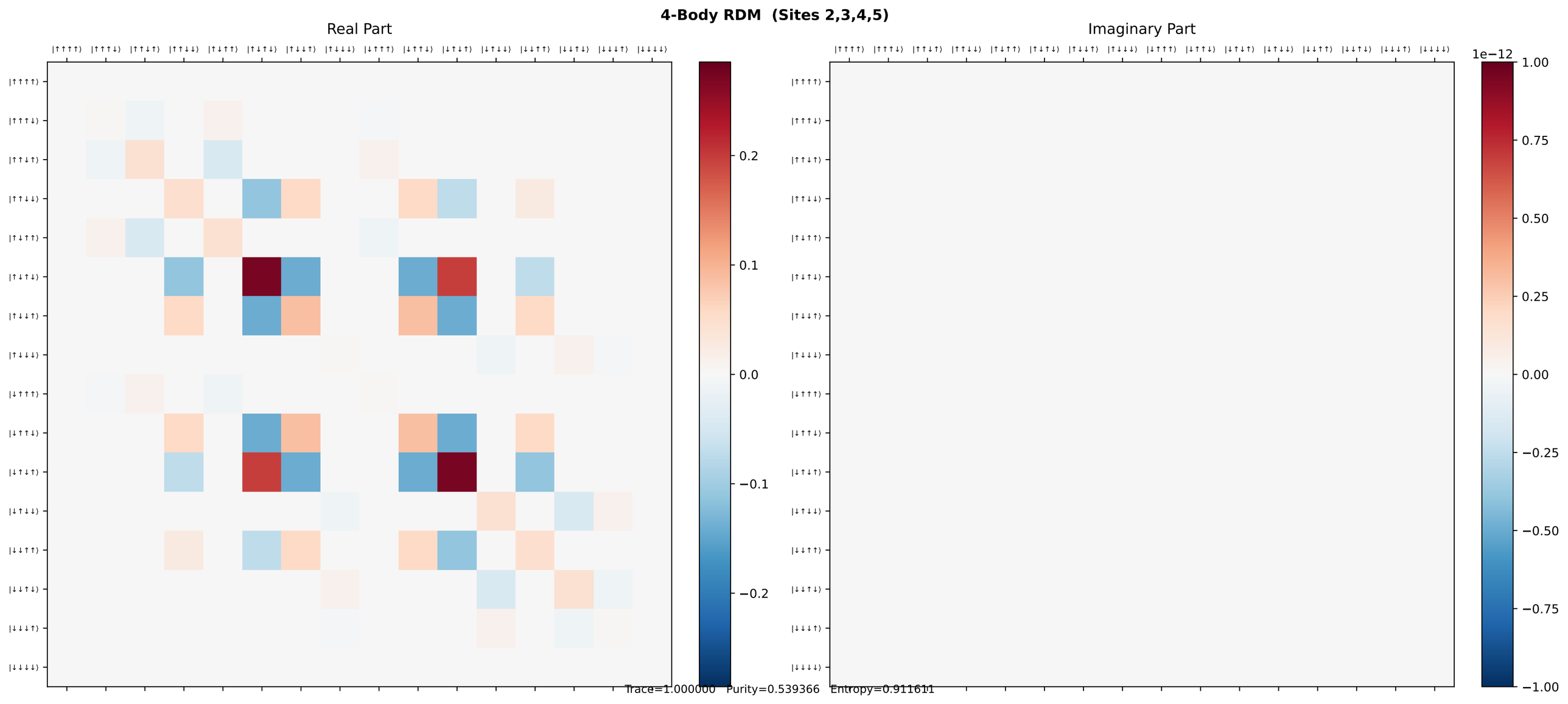
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 2,3,4,5)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.539366 Entropy=0.911611

```
RDM Metrics  (Sites = 2,3,4,5)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5393659101
  Von Neumann entropy    : 0.9116106080
  Eigenvalue  min        : -5.529774e-17
  Eigenvalue  max        : 7.159102e-01
```

GME Analysis — 4-Body RDM (Sites 2,3,4,5)

Bipartitions (Sites 2,3,4,5)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.53936591$

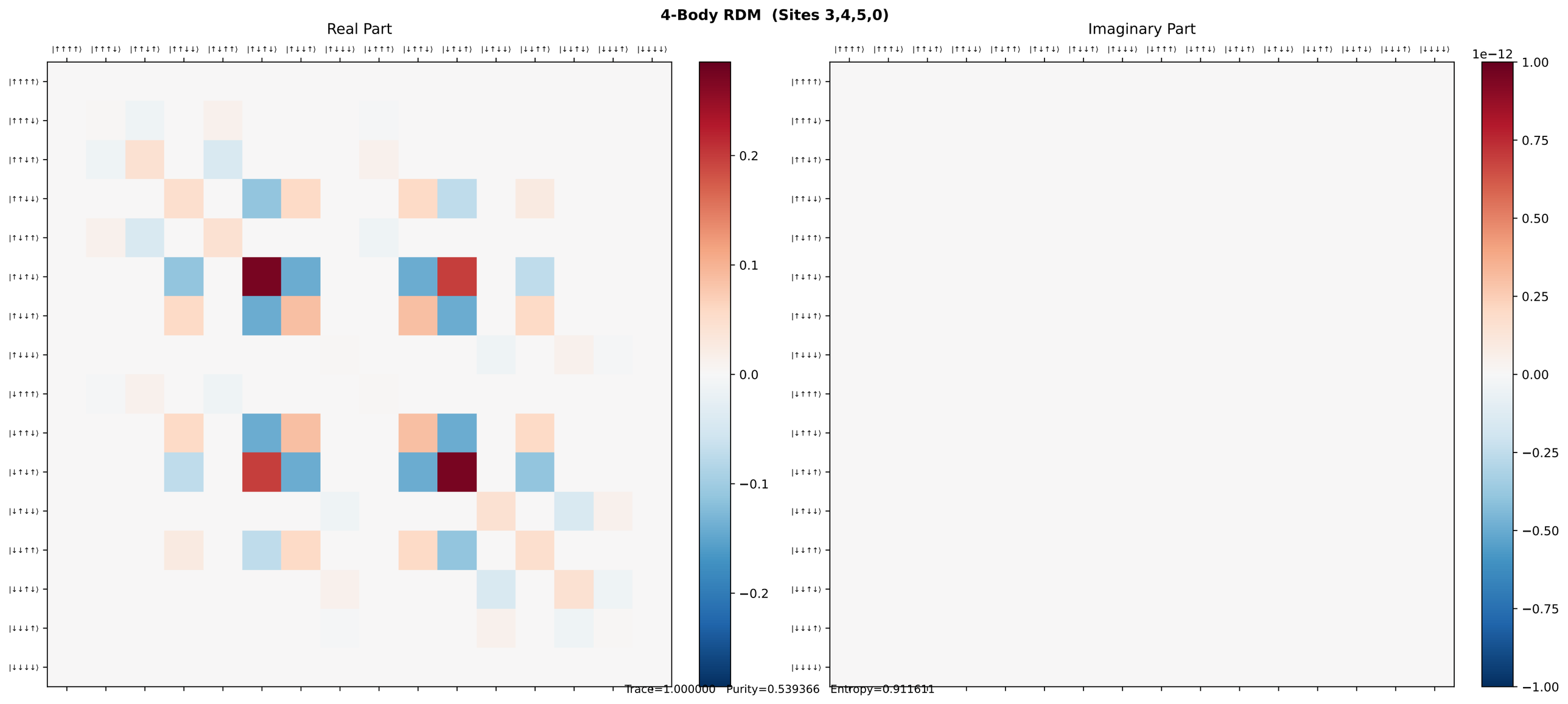
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 3,4,5,0)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.539366 Entropy=0.911611

```
RDM Metrics  (Sites = 3,4,5,0)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5393659101
  Von Neumann entropy    : 0.9116106080
  Eigenvalue  min        : -9.106885e-17
  Eigenvalue  max        : 7.159102e-01
```

GME Analysis — 4-Body RDM (Sites 3,4,5,0)

Bipartitions (Sites 3,4,5,0)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.53936591$

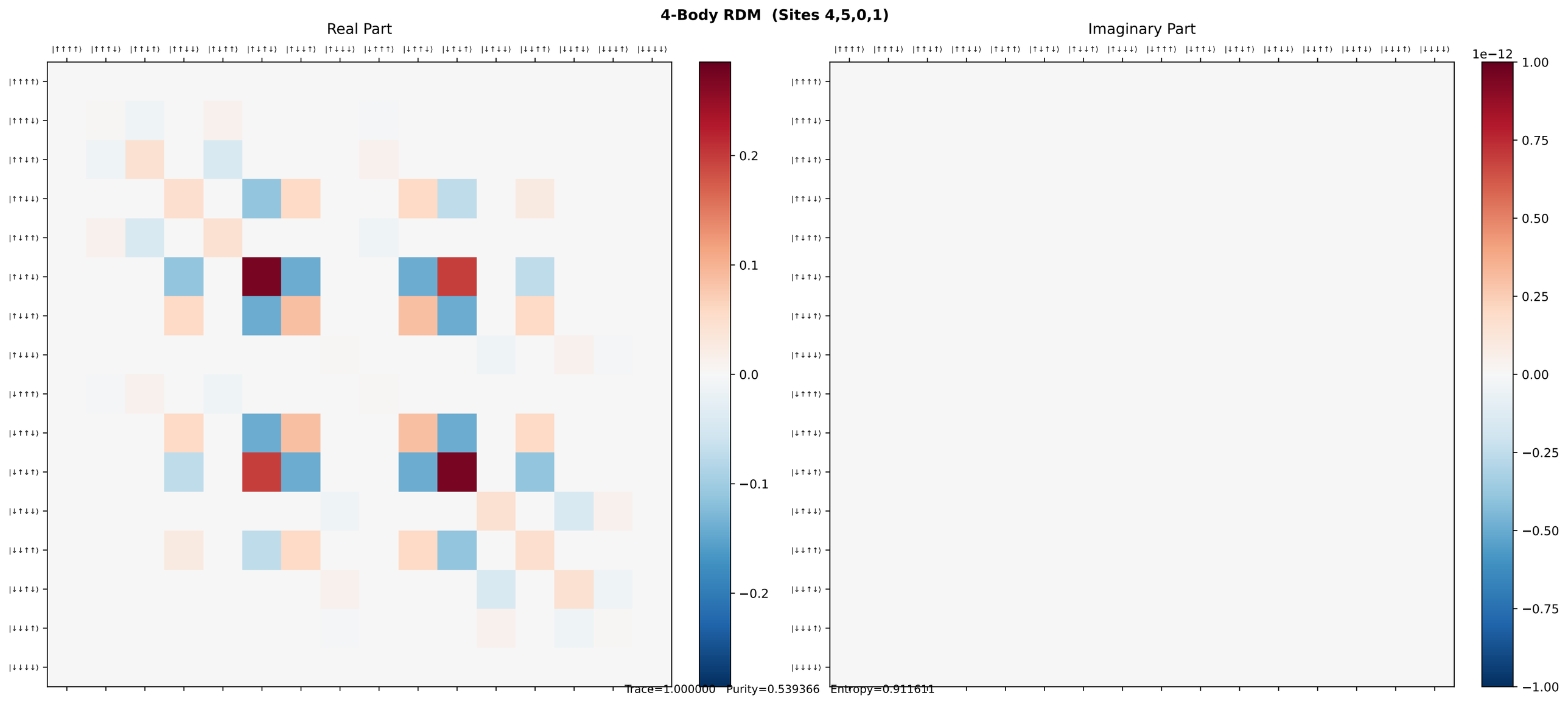
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 4,5,0,1)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.539366 Entropy=0.911611

```
RDM Metrics  (Sites = 4,5,0,1)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5393659101
  Von Neumann entropy    : 0.9116106080
  Eigenvalue  min        : -1.706147e-17
  Eigenvalue  max        : 7.159102e-01
```

GME Analysis — 4-Body RDM (Sites 4,5,0,1)

Bipartitions (Sites 4,5,0,1)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.53936591$

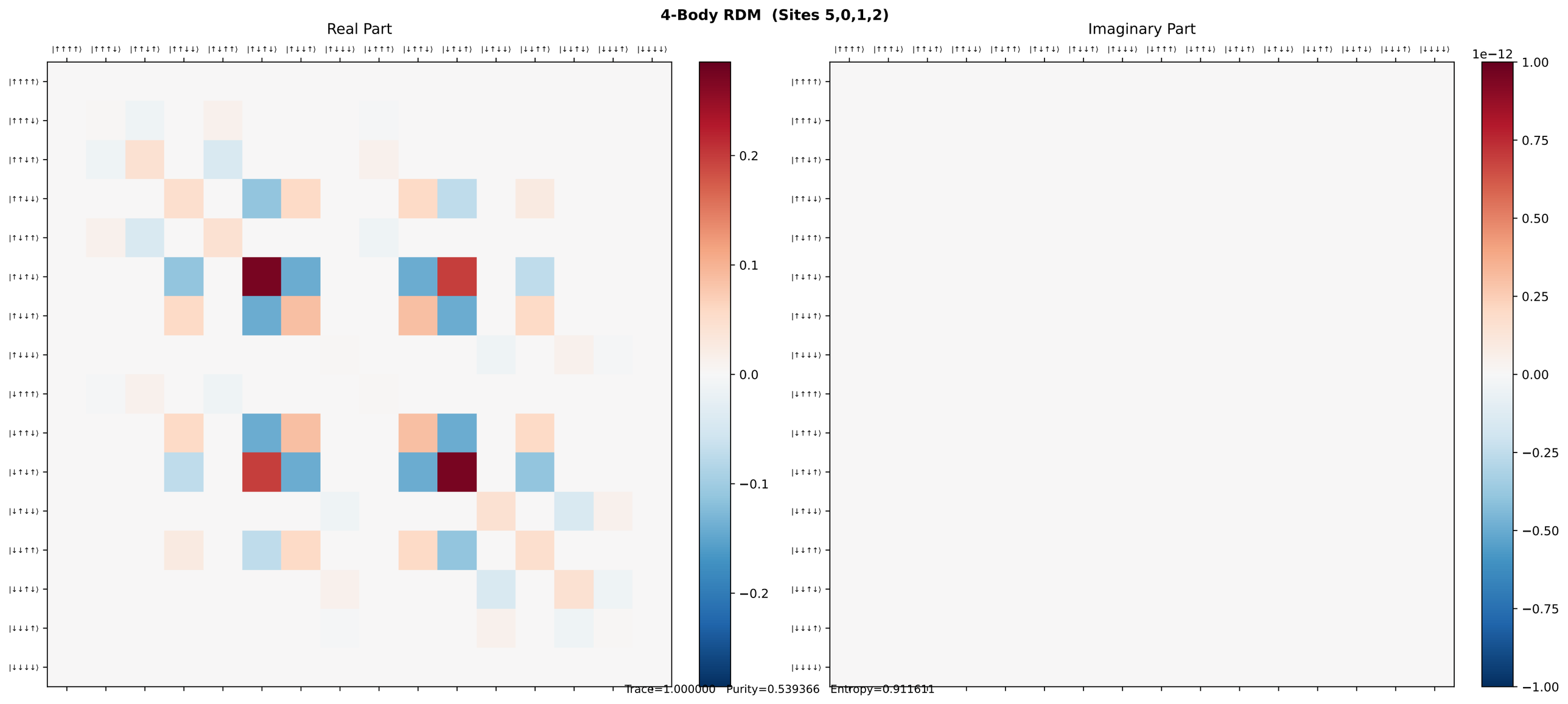
A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).


4-Body RDM (Sites 5,0,1,2)
Real Part
Imaginary Part
1e−12
Trace=1.000000 Purity=0.539366 Entropy=0.911611

```
RDM Metrics  (Sites = 5,0,1,2)
=========================================================

  Trace(ρ)               : 1.0000000000
  Purity Tr(ρ²)          : 0.5393659101
  Von Neumann entropy    : 0.9116106080
  Eigenvalue  min        : -2.534680e-17
  Eigenvalue  max        : 7.159102e-01
```

GME Analysis — 4-Body RDM (Sites 5,0,1,2)

Bipartitions (Sites 5,0,1,2)

Global purity $\gamma = \mathrm{Tr}(\rho^2) = 0.53936591$

A	B	$\mathrm{Tr}(\rho_A^2)$

STATUS: MIXED STATE

The extracted n-spin RDM is mixed ($\mathrm{Tr}(\rho^2) < 1$).

The pure-state C_GME formula is not applicable here.

Convex-roof extension required (planned for next phase).